

Supplement: Deriving the Secular Equation for H_2^+

Complete step-by-step derivation of the secular determinant and eigenvalue equations

1. The Variational Principle & Linear Combinations of Atomic Orbitals (LCAO)

We construct molecular orbitals as linear combinations of hydrogen 1s atomic orbitals centered on nuclei A and B:

$$\psi_{\text{MO}} = c_A \phi_A + c_B \phi_B$$

where ϕ_A and ϕ_B are normalized hydrogen 1s wavefunctions. The variational principle states that any trial wavefunction yields an energy greater than or equal to the true ground state energy: $E[\psi] \geq E_0$. To find the best linear combination, we minimize E with respect to the coefficients c_A and c_B .

2. Setting Up the Energy Functional

The expectation value of the Hamiltonian operator is:

$$E(c_A, c_B) = \langle \psi | H | \psi \rangle / \langle \psi | \psi \rangle$$

Expanding $\langle \psi | H | \psi \rangle$:

$$\langle \psi | H | \psi \rangle = c_A^2 H_{AA} + c_B^2 H_{BB} + 2c_A c_B H_{AB}$$

where $H_{ij} = \langle \phi_i | H | \phi_j \rangle$ are the Hamiltonian matrix elements. Similarly, $\langle \psi | \psi \rangle$ expands to:

$$\langle \psi | \psi \rangle = c_A^2 + c_B^2 + 2c_A c_B S_{AB}$$

where $S_{AB} = \langle \phi_A | \phi_B \rangle$ is the overlap integral. By symmetry, $H_{AA} = H_{BB} = \alpha$ and $S_{AA} = S_{BB} = 1$.

3. Minimization Conditions: The Secular Equations

To minimize E , we set the partial derivatives with respect to c_A and c_B to zero:

$$dE/dc_A = 0 \text{ and } dE/dc_B = 0$$

Applying the quotient rule and simplifying, these conditions yield:

$$(H_{AA} - E)c_A + (H_{AB} - E \cdot S_{AB})c_B = 0$$

$$(H_{AB} - E \cdot S_{AB})c_A + (H_{BB} - E)c_B = 0$$

These are the **secular equations** in matrix form. For a nontrivial solution (c_A, c_B not both zero), the determinant of the coefficient matrix must vanish:

4. The Secular Determinant

$$\det(H_{ij} - E \cdot S_{ij}) = 0$$

Explicitly:

$$\begin{vmatrix} H_{AA} - E & H_{AB} - E \cdot S_{AB} \\ H_{AB} - E \cdot S_{AB} & H_{BB} - E \end{vmatrix} = 0$$

By symmetry ($H_{AA} = H_{BB} = \alpha$ and $S_{AA} = S_{BB} = 1$), this simplifies to:

$$(\alpha - E)^2 - (\beta - E \cdot S_{AB})^2 = 0$$

where $H_{AB} = \beta$ for convenience.

5. Solving for the Eigenvalues

Expanding the determinant and factoring:

$$[(\alpha - E) + (\beta - E \cdot S_{AB})][(\alpha - E) - (\beta - E \cdot S_{AB})] = 0$$

This gives two solutions:

Bonding orbital (sigma):

$$E_+ = (\alpha + \beta) / (1 + S_{AB})$$

Since beta is negative (stabilizing), $E_+ < \alpha$, giving lower energy.

Antibonding orbital (sigma*):

$$E_- = (\alpha - \beta) / (1 - S_{AB})$$

Here $E_- > \alpha$, giving higher energy. Note that the antibonding orbital is raised more than the bonding orbital is lowered because $(1 - S_{AB}) < (1 + S_{AB})$.

6. Finding the Molecular Orbital Coefficients

For each eigenvalue, we solve the secular equations:

For E_+ (bonding): $c_A = c_B$

$$\psi_+ = (1/\sqrt{2(1 + S_{AB})}) (\phi_A + \phi_B)$$

Constructive interference in the internuclear region — electron density accumulates between the nuclei.

For E_- (antibonding): $c_A = -c_B$

$$\psi_- = (1/\sqrt{2(1 - S_{AB})}) (\phi_A - \phi_B)$$

Destructive interference creates a **node** between the nuclei, depleting electron density from the bonding region.

Summary: Key Results

The secular equation analysis yields molecular orbital energies that depend crucially on the overlap integral S_{AB} and the resonance integral $\beta = H_{AB}$. The bonding orbital is stabilized and the antibonding orbital is destabilized, with the asymmetry arising from the overlap integral in the denominators. This framework explains why H_2^+ is stable while He_2 is not.